D. Hudson Smith | curriculum vitae

dane2@clemson.edu • dhudsmith • dhudsmith

Research Vision

I develop techniques for incorporating prior knowledge into machine learning systems for data-constrained applications. This synthesis of established domain knowledge with flexible machine learning methods leads to better anomaly detection rates, improved sample efficiency, and explainable inference, without sacrificing the expressive power of data-driven techniques like deep learning.

Positions

Assistant Professor — **Clemson University**
School of Mathematical and Statistical Sciences — *8/2024 – present*

Co-Lead — **Clemson University**
Clemson/MUSC Artificial Intelligence Hub — *7/2022 – present*

Director of Applied Machine Learning — **Clemson University**
Research Computing and Data, CCIT — *1/2023 – 7/2024*

Machine Learning Research Associate — **Clemson University**
Watt Family Innovation Center — *3/2018 – 12/2022*

Data Scientist — **Columbus, OH**
Dynamit Technologies — *12/2016 – 3/2018*

Education

Ohio State University — **Columbus, OH**
Ph.D. in Theoretical Atomic Physics, GPA: 3.9/4.0 — *2011-2016*
Research: theoretical atomic physics, quantum field theory, computer simulation of quantum systems
Thesis: Inducing Resonant Interactions in Ultracold Atoms with an Oscillating Magnetic Field
Advisor: Dr. Eric Braaten, `braaten.1@osu.edu`

Erskine College — **Due West, SC**
B. S. in Physics, B. A. in Mathematics, GPA: 3.9/4.0 — *2007-2011*

Publications

Machine Learning and Applied Research

1. Raza, A. & Smith, D. H. Massively Parallel Exact Inference for Hawkes Processes. In *Advances in Neural Information Processing Systems* To appear (2026). `https://arxiv.org/abs/2604.01342`.
2. Bakumenko, A., Smith, D. H. & Hoelscher, J. Early-Horizon Multimodal ICU Mortality Prediction Without Retraining. In *Proceedings of the 11th Machine Learning for Healthcare Conference* **340,** 214–240 (PMLR, 2026). `https://proceedings.mlr.press/v340/bakumenko26a.html`.
3. Floyd, S. B., Almeldien, A. G., Smith, D. H., Judkins, B., Krohn, C. E., Reynolds, Z. C., Jeray, K. & Obeid, J. S. Using artificial intelligence to develop a measure of orthopaedic treatment success from clinical notes. *Frontiers in Digital Health* **7,** 1523953. `https://doi.org/10.3389/fdgth.2025.1523953` (Apr. 2025).

4. Formby, B., Wang, K.-C. & Smith, D. H. Attention-Based Multiple Instance Learning for Achilles Tendinopathy Detection in Ultrasound Imaging. In *2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)* 1–5 (Apr. 2025). https://doi.org/10.1109/ISBI60581.2025.10981252.

5. Walker, E. B., Bridges, A. W. & Smith, D. H. A color comparison between inkjet and dye sublimation on non-paper substrates after environmental exposure. *The Journal of Technology, Management, and Applied Engineering* **40.** https://doi.org/10.31274/jtmae.16889 (Jan. 2025).

6. Cull, J., Morrow, D., Manasco, C., Vaughan, A., Eicken, J. & Smith, H. A quality assessment tool for focused abdominal sonography for trauma examinations using artificial intelligence. *Journal of Trauma and Acute Care Surgery* **98,** 111–116. https://doi.org/10.1097/TA.0000000000004425 (Jan. 2025).

7. Smith, D. H., Ehrett, C. & Warren, P. Unsupervised detection of coordinated information operations in the wild. *EPJ Data Science* **14,** 26. https://doi.org/10.1140/epjds/s13688-025-00544-y (Mar. 2025).

8. Smith, D. H., Nisbet, N., Ehrett, C., Tica, C. I., Atwell, M. M. & Weisensee, K. E. Modeling human decomposition: A Bayesian approach. *Forensic Science International* **367,** 112309. https://doi.org/10.1016/j.forsciint.2024.112309 (2025).

9. Fine, J. A., Smith, D. H., Oliveira, C., Deas, N., Shellnutt, S., Stotzky, R. & Clyburn, R. How negativity and policy content drive the spread of political messages. *Journal of Information Technology & Politics* **22,** 296–305 (2025).

10. Antao, A., Burton, J. D., Dawson, D., Gemmill, J., Gerstener, Z., Godfrey, B., Groel, S., Jordan, Z., Ligon, B., Smith, D., Moersen, M., Nandigam, S., Rengier, N., Ryans, D., Smith, H. & Wright, T. Modernizing Clemson University's Palmetto Cluster: Lessons Learned from 17 Years of HPC Administration. In *Practice and Experience in Advanced Research Computing 2024: Human Powered Computing (PEARC '24)* 1–9 (July 2024). https://doi.org/10.1145/3626203.3670543.

11. Weisensee, K. E., Tica, C. I., Atwell, M. M., Ehrett, C., Smith, D. H., Carbajales-Dale, P., Claflin, P. & Nisbet, N. geoFOR: a collaborative forensic taphonomy database for estimating the postmortem interval. *Forensic Science International,* 111934. https://doi.org/10.1016/j.forsciint.2024.111934 (2024).

12. Cranmer, G. A., Linvill, D., Smith, H., Denham, B., Bober, J., Nutt, K. & Seaton, W. Social media trolls as faux third-party agents of image repair: China's disinformation campaign and statecraft in the Daryl Morey affair. *Journal of Applied Communication Research* **52,** 5–26. https://doi.org/10.1080/00909882.2023.2282508 (2024).

13. Smith, D. H., Lineberger, J. P. & Baker, G. H. On the Relevance of Temporal Features for Medical Ultrasound Video Recognition. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* 744–753 (2023).

14. Erno, J., Gomes, T., Baltimore, C., Lineberger, J. P., Smith, D. H. & Baker, G. H. Automated Identification of Patent Ductus Arteriosus Using a Computer Vision Model. *Journal of Ultrasound in Medicine* (2023).

15. Smith, D. H. & Walker, E. B. Real-world evaluation of artificial intelligence-based color corrections for social media content creators. *Journal of Print and Media Technology Research* **12,** 15–20 (2023).

16. Smith, D. H., Ehrett, C., Weisensee, K. & Tica, C. Commentary on: Megyesi MS, Nawrocki SP, Haskell NH. Using accumulated degree-days to estimate the postmortem interval from decomposed human remains. *Journal of Forensic Sciences* **68,** 355–358 (2023).

17. Taye, M., Morrow, D., Cull, J., Smith, D. H. & Hagan, M. Deep Learning for FAST Quality Assessment. *Journal of Ultrasound in Medicine* (2022).

18. Walker, E. & Smith, H. ColorNet Case Study: Impacts on Undergraduate Students Participating in a Small Multidisciplinary Research Course. In *INTED2022 Proceedings* 1080–1084 (2022).

19. Woo, M., Mishra, P., Lin, J., Kar, S., Deas, N., Linduff, C., Niu, S., Yang, Y., McClendon, J., Smith, D. H., *et al.* Complete and Resilient Documentation for Operational Medical Environments Leveraging Mobile Hands-free Technology in a Systems Approach: Experimental Study. *JMIR mHealth and uHealth* **9,** e32301 (2021).
20. Kar, S., Mishra, P., Lin, J., Woo, M.-J., Deas, N., Linduff, C., Niu, S., Yang, Y., McClendon, J., Smith, D. H., *et al.* Systematic Evaluation and Enhancement of Speech Recognition in Operational Medical Environments. In *2021 International Joint Conference on Neural Networks* (*IJCNN*) 1–8 (2021).
21. Ehrett, C., Linvill, D. L., Smith, H., Warren, P. L., Bellamy, L., Moawad, M., Moran, O. & Moody, M. Inauthentic Newsfeeds and Agenda Setting in a Coordinated Inauthentic Information Operation. *Social Science Computer Review* (June 2021).
22. Mayes, E., Lineberger, J. P., Mayer, M., Sanborn, A., Smith, H. & Walker, E. Automated Brand Color Accuracy for Realtime Video. *SMPTE Motion Imaging Journal* **130,** 45–49 (2021).
23. Mayer, M., Walker, E. B. & Smith, D. H. ColorNet 1.5: Improving machine learning color correction for video. In *Proceedings of the 2021 NAB Broadcast Engineering and IT Conference* (2021).
24. Freeman, D., Gupta, S., Smith, D. H., Maja, J. M., Robbins, J., Owen, J. S., Peña, J. M. & de Castro, A. I. Watson on the Farm: Using Cloud-Based Artificial Intelligence to Identify Early Indicators of Water Stress. *Remote Sensing* **11,** 2645 (Nov. 2019).

Preprints

1. Smith, D. H. Variance Fingerprints: Component-Wise Identification of Content Factors by Augmentation Design. *arXiv preprint arXiv:2610.14260.* `https://arxiv.org/abs/2610.14260` (2026).
2. Smith, D. H. & Raza, A. Compression Hurts, Pooling Helps: Information Loss in Rayleigh-Scale Estimation from B-Mode Ultrasound. *arXiv preprint arXiv:2609.19525.* `https://arxiv.org/abs/2609.19525` (2026).
3. Formby, B., Wang, K.-C. & Smith, D. H. Spatial Attention Noise Masking for Causally Sufficient Interpretability. *arXiv preprint arXiv:2608.14725.* `https://doi.org/10.48550/arXiv.2608.14725` (Aug. 2026).
4. Hart, J., Smith, H., Rennert, L. & McMahan, C. S. Infectious Disease Forecasting via Physics-Informed Machine Learning. *bioRxiv preprint.* `https://www.biorxiv.org/content/10.64898/2026.06.12.731957v1` (June 2026).
5. Bakumenko, A., Hoelscher, J., Smith, D. H., Bakumenko, S., Masino, A. J. & Obeid, J. S. Multimodal ICU in-hospital mortality prediction: A scoping review of transparency, calibration, and translation readiness. *Research Square preprint.* `https://doi.org/10.21203/rs.3.rs-9217081/v1` (Mar. 2026).
6. Bakumenko, A., Hoelscher, J. & Smith, D. H. Transparent Early ICU Mortality Prediction with Clinical Transformer and Per-Case Modality Attribution. *arXiv preprint arXiv:2511.15847.* `https://doi.org/10.48550/arXiv.2511.15847` (Nov. 2025).

Physics

1. Smith, D. H. & Volosniev, A. G. Engineering momentum profiles of cold-atom beams. *Physical Review A* **100,** 033604 (2019).
2. Volosniev, A. & Smith, D. H. Impenetrability in Floquet Scattering in One Dimension. *Few-Body Systems* **59,** 1–9 (2018).
3. Smith, D. H. Inducing Resonant Interactions in Ultracold Atoms with a Modulated Magnetic Field. *Physical Review Letters* **115,** 193002 (2015).
4. Langmack, C., Smith, D. H. & Braaten, E. Association of atoms into universal dimers using an oscillating magnetic field. *Physical Review Letters* **114,** 103002 (2015).

5. Braaten, E., Langmack, C. & Smith, D. H. Born-Oppenheimer approximation for the XYZ mesons. *Physical Review D* **90,** 014044 (2014).
6. Braaten, E., Langmack, C. & Smith, D. H. Selection Rules for Hadronic Transitions of XYZ Mesons. *Physical Review Letters* **112,** 222001 (2014).
7. Smith, D. H., Braaten, E., Kang, D. & Platter, L. Two-body and three-body contacts for identical bosons near unitarity. *Physical Review Letters* **112,** 110402 (2014).
8. Langmack, C., Smith, D. H. & Braaten, E. Atom Loss Resonances in a Bose-Einstein Condensate. *Physical Review Letters* **111,** 023003 (2013).
9. Langmack, C., Smith, D. H. & Braaten, E. Avalanche mechanism for the enhanced loss of ultracold atoms. *Physical Review A* **87,** 023620 (2013).
10. Langmack, C., Smith, D. H. & Braaten, E. Avalanche mechanism for atom loss near an atom-dimer Efimov resonance. *Physical Review A* **86,** 022718 (2012).

Patent

2022: E. B. Walker and D. Hudson Smith. *Color adjustment system for disparate displays*. U.S. Patent 11,501,472, granted November 15, 2022.

Research awards

Amounts are total project awards, not individual funding shares.

2026–2027: Clemson School of Health Research *$50k*

(PI) FAST Pilot Data Collection for AI-Enabled Guidance

from 2026: NSF STRIDE, two awards *$199k combined*

(Co-I) Forensic taphonomy database; AI-augmented X-ray fracture tool

1/2025–12/2025: Clemson Media Forensics Hub *$15k*

(PI) Modeling the Tactics of State-Backed Social Media Information Operations

12/2024–7/2025: Booz Allen Hamilton *$511,384*

(PI) Foreign Influence Network Detection Development

9/2024–8/2026: CDC *$2.8MM*

(Senior Personnel) Disease Monitoring Infrastructure to support Detection, Surveillance, and Forecasting of Emerging and Re-Emerging Pathogens (DMI-DEEP)

5/2024–4/2026: NIH *$405,677*

(Senior Personnel) Understanding and Visualizing Emergency Department Clinician Well-Being and Strain

1/2024: NIH COBRE Pilot Award *$345,520*

(Co-PI) Team Science Supplement: SC COBRE for Translational Research Improving Musculoskeletal Health (SC TRIMH)

9/2023: CDC Center for Forecasting and Outbreak Analytics *$7MM*

(Senior Personnel) Disease Modeling and Analytics to inform Outbreak Preparedness, Response, Intervention, Mitigation, and Elimination in South Carolina (DMA-PRIME)

8/2023: NIH COBRE Renewal Award *$11,113,053*

(Co-PI) COBRE Renewal: Translational Research Improving Musculoskeletal Health (TRIMH) (SC TRIMH)

7/2023: Clemson/MUSC AI Hub Augmentation Award *$25k*

(Co-PI) Automated Identification of Patent Ductus Arteriosus using a Computer Vision Model

4/2023: Clemson University Research Foundation Technology Maturation Award *$20k*

(Senior Personnel) Integration of Smart Needle Measurements and Functional Ultrasonography by Machine Learning for Quantitative Monitoring and Assessment of Acupuncture Myofascial Pain Management

1/2022: Clemson-MUSC Artificial Intelligence Hub *$5k*

(Fellow) Artificial Intelligence Advocate

7/2021: RHBSSI Seed Grant (Clemson University) *$45k*

(Co-PI) ColorNet: Developing AI-based color correction tools for sports media applications

6/2021: ACRE Competitive Grants Program (SCDA) *$120k*

(PI) AI Master Gardener for Greenhouse Supervision

4/2021: Prisma Health Seed Grant *$20k*

(Senior Personnel) Automated Quality Assessment of FAST Exams

2/2021: ACRE Competitive Grants Program (SCDA) *$20k*

(Co-PI) AI for Fruit and Vegetable Harvesting in South Carolina

2/2021: CU Seed Grant, Tier 1 (Clemson University) *$5k*

(Co-PI) ColorNet: An AI-based color management system for live video

11/2019: CURF Tech Maturation Fund (Clemson) *$29k*

(Co-PI) ColorNet: Consistent display of Clemson brand colors using artificial intelligence

8/2019: Erwin Center for Brand Communications (Clemson University) *$8k*

(Co-PI) AI for on the fly color correction of sports footage

7/2018: ACRE Competitive Grants Program (SCDA) *$105k*

(Co-PI) Rapid Chicken Sex Determination with Multiple Mechanisms and AI

Teaching Experience

Mentorship Experience

Graduate students: I advise or co-advise doctoral and master's students in machine learning and applied research. My mentoring has included students in mathematical and statistical sciences, biomedical data science and informatics, electrical and computer engineering, computing, and orthodontics.

Undergraduates: In my 5 years leading the Watt AI program, I worked with over 100 Creative Inquiry students applying Machine Learning techniques to research projects at Clemson.

Interns: Since 2018, I have mentored more than 15 undergraduate student interns. These interns worked on machine-learning related projects in diverse disciplines.

Clemson University

Graduate courses: Developed and taught Introduction to Machine Learning I and II (MATH 8710/8720), including original problem sets and exams.

Fall 2024: Instructor for Introduction to Data Science (MATH 2800).

Spring 2023–present: Instructor for Data Science, Machine Learning, and Deep Learning Workshops for Research Computing and Data. Selected topics include "Data Visualization in Python", "Introduction to Deep Learning with PyTorch", and "Attention, Transformers, and LLMs: a hands-on introduction in PyTorch"

Spring 2018–Fall 2022: Instructor for Watt AI Creative Inquiry course for 9 consecutive semesters

Spring 2020–Fall 2021: Designed intro to artificial intelligence curriculum for undergraduates from diverse majors

Fall 2021–Fall 2022: Led weekly journal club with advanced students

Fall 2018–Spring 2019: Instructor for Ulbrich CI focused on manufacturing analytics

Ohio State University

Fall 2015: Tutor for graduate level classical mechanics course

Fall 2012–Spring 2013: Recitation and lab instructor for Physics: Vibrations, Fluids, Thermodynamics, and Special Relativity

Erskine College

Spring 2010: Lab instructor for Modern Physics

Fall 2009: Teaching assistant for Calculus

Fall 2008–Fall 2009: Teaching assistant for Introductory Physics

Fall 2008–Fall 2010: Writing assistant for various subjects

Presentations and Conference Abstracts

Machine learning

1. Gearhart, A., Elrod, M., Gomes, T., Golbus, A., Baltimore, C., Wakser, C., Graham, J., Lai, W., Powell, A., Smith, D. & Baker, G. Abstract 4135928: Externally Validated Deep Learning Model for Patent Ductus Arteriosus Detection by Echocardiography in Preterm Infants. *Circulation* **150.** Meeting abstract, A4135928. `https://www.ahajournals.org/doi/10.1161/circ.150.suppl_1.4135928` (2024).
2. Smith, D. H. & Lineberger, J. P. *On the Relevance of Temporal Features for Medical Ultrasound Video Recognition* International Conference on Medical Image Computing and Computer Assisted Intervention. Oct. 2023.
3. Gemmill, J. & Smith, D. H. *Member Panel on AI: Supporting AI in an HPC Environment* Coalition for Academic Scientific Computation Fall Meeting. Oct. 2023.
4. Smith, D. H. *Artificial Intelligence: from Duck Digestion to Conversational Companions* Clemson University Data Science Conference. May 2023.
5. Weisensee, K., Tica, I., Smith, D. H., Ehrett, C. & Carbajales-Dale, P. *Postmortem Interval Estimation Using a Novel Data set and Methods* Annual Meeting of the American Academy of Forensic Sciences. Feb. 2023.
6. Smith, D. H. *Quality Assessment of FAST Exams using Deep Learning* Clemson/MUSC AI Hub Seminar. Nov. 2021.
7. Smith, D. H. SMRF: a Cloud-Based Social Media Research Framework. In *Research Running on Cloud Compute & Emerging Technologies* **2021,** 11–12 (Apr. 2021).
8. Walker, E. B., Smith, D. H., Lineberger, J. P., Mayer, M. L., Mayes, E. E. & Sanborn, A. T. 67-3: ColorNet: A Neural Network-Based System for Consistent Display of Brand Colors for Video. In *SID Symposium Digest of Technical Papers* **51,** 1001–1004 (2020).
9. Fine, J., Deas, N., Shellnut, S., Sargent, J. & Smith, D. H. Content Analyzing Political Tweets using Natural Language Processing: Opportunities and Challenges. In *Southern Political Science Association Conference* (2019).
10. Zhang, T., Moody, M., Nelon, J. P., Boyer, D. M., Smith, D. H. & Visser, R. D. Using Natural Language Processing to Accelerate Deep Analysis of Open-Ended Survey Data. In *2019 SoutheastCon* 1–3 (2019).

Physics

1. Mohapatra, A., Smith, D. H. & Braaten, E. Dissociation of Cooper pairs in the BCS Limit using an Oscillating Magnetic Field. In *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts* (2016).
2. Smith, D. H. Induced two-body scattering resonances from a square-well potential with oscillating depth. In *EPJ Web of Conferences* **113,** 02005 (2016).

3. Braaten, E., Mohaptra, A. & Smith, D. H. Initial Atom Loss Rate after the Sudden Ramp of a BEC to Unitarity. In *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts* (2016).
4. Smith, D. H. Inducing Resonant Interactions in Ultracold Atoms with an Oscillating Magnetic Field. In *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts* **1,** 3010 (2015).
5. Smith, D., Langmack, C., Braaten, E., *et al.* Avalanche Mechanism for the Enhanced Loss of Ultracold Atoms. In *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts* **1,** 5007P (2013).
6. Braaten, E. & Smith, D. Avalanche Mechanism for Multiple Atom Loss near an Efimov Atom-Dimer Resonance. In *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts* (2012).

Professional Service

2026: NeurIPS reviewer, four papers.

2025: MICCAI reviewer, six papers.

2022–present: Co-lead, Clemson/MUSC Artificial Intelligence Hub; organizes the annual AI Hub Summit and AI project consultation service.

Computational tools

- Python, R, SQL, bash, C++, C#, Java, LaTeX
- Deep Learning and Probabilistic Programming: PyTorch, Pyro, NumPyro, JAX
- Experience with Cloud and cluster computing environments
- Hardware-accelerated array programming for scientific computing

Honors and Awards

2024: National Communication Association Bill Eadie Distinguished Scholarly Article Award, for co-authored article on social media trolls and image repair in the Daryl Morey affair.

2016: Presidential Fellow, OSU

2013: Winner of Physics Dept. Poster Competition, OSU

2011: Fowler Fellow, OSU

2011: University Fellow, OSU

2010: T. Kincannon Mathematics Award, Erskine College

2010: Junkin Physics Award, Erskine College

2008: Garnet Circle Award, Erskine College

2007: Roy M. Smith Mathematics Scholarship, Erskine College